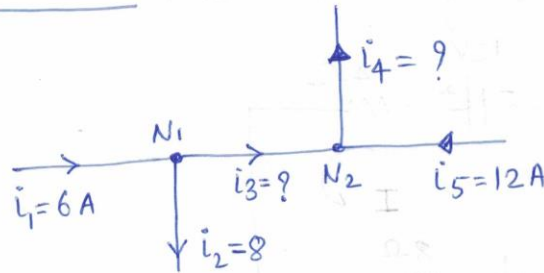


Module 2 :-

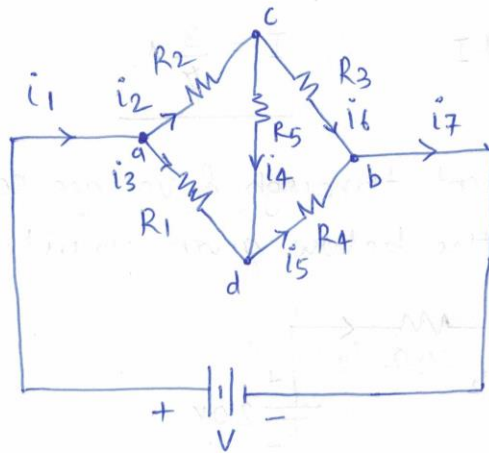
* Problems on KCL :-

①



$i_3 = -2A$ (leftward), $i_4 = 10A$ (upward).

②



For the given circuit,
if $i_1 = 12A$, $i_2 = 7A$ &
 $i_5 = 5A$, then find
other currents.

→ at node a, using KCL,

$$i_1 = i_2 + i_3$$

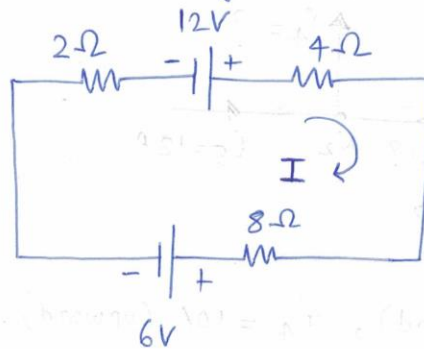
$$i_3 = i_1 - i_2 = 5A.$$

at node d, $i_5 = i_3 + i_4 \therefore i_4 = i_5 - i_3 = 0A.$

$i_6 = i_2 = 7A$, $i_7 = i_1 = 12A.$

KVL :-

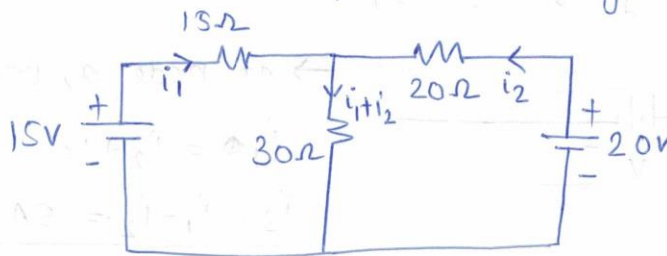
① Find current I flowing in the circuit.



using KVL, $+12 - 4I - 8I - 6 - 2I = 0$

$$6 = 14I \quad \therefore I = \frac{3}{7} \text{ A}$$

② using KVL, find current through & voltage across 30Ω resistor of the below given circuit.



using KVL for left portion of circuit.

$$15 - 15i_1 - 30(i_1 + i_2) = 0 \rightarrow 15 = 45i_1 + 30i_2 \quad \text{--- ①}$$

Similarly, for right portion/loop of the circuit,

$$20 - 20i_2 - 30(i_1 + i_2) = 0 \rightarrow 20 = 50i_2 + 30i_1 \quad \text{--- ②}$$

$$\text{③} - \text{①} - \rightarrow 30 = 75i_2 + 45i_1 \quad \text{③}$$

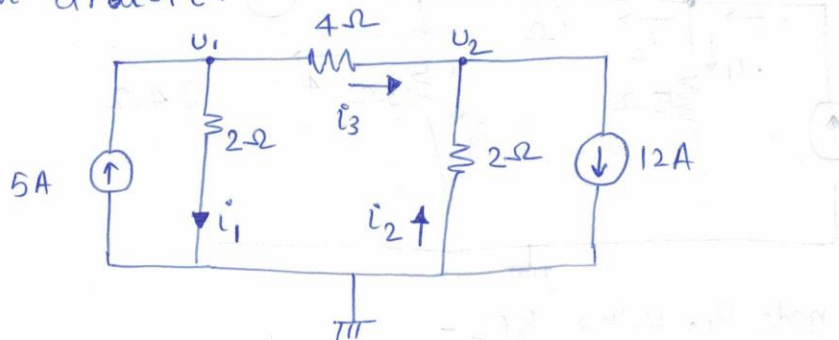
$$15 = 45i_2 \rightarrow i_2 = \frac{1}{3} \text{ A}$$

$$i_1 = \frac{1}{9} \text{ A}$$

Net current in $30 = i_1 + i_2 = \frac{4}{9} \text{ A}$. voltage drop = $\frac{40}{9} \text{ V}$.

Nodal Analysis :-

- ① Calculate node voltages & branch currents for the below given circuit.



Sol:-

at node v_1 - using KCL -

$$5 = i_1 + i_3 = \frac{v_1}{2} + \frac{v_1 - v_2}{4}$$

$$6v_1 - 2v_2 = 40 \rightarrow 20 = 3v_1 - v_2 \quad \text{--- (1)}$$

at node v_2 , using KCL -

$$i_2 + i_3 = 12 \rightarrow \frac{0 - v_2}{2} + \frac{v_1 - v_2}{4} = 12$$

$$2v_1 - 6v_2 = 96 \rightarrow v_1 - 3v_2 = 48 \quad \text{--- (2)}$$

Solving ① & ② -

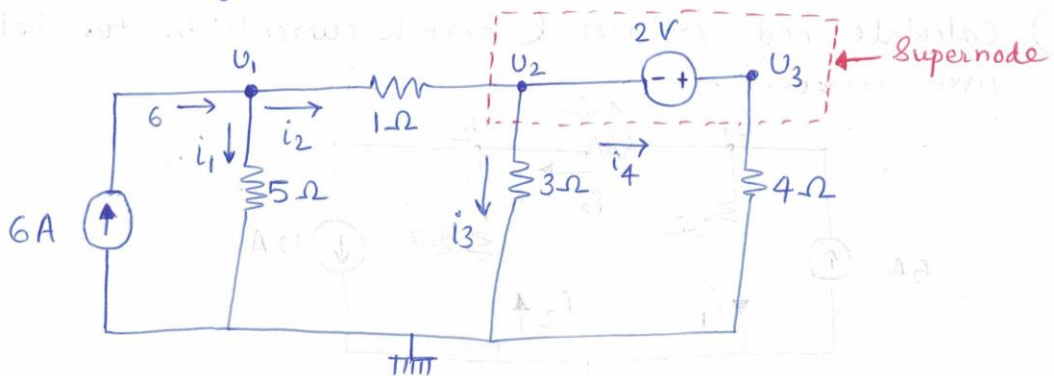
$$-8v_2 = 124, \quad \underline{v_2 = -15.5V}$$

$$\underline{v_1 = 48 + 3v_2 = 1.5V.}$$

$$\therefore i_1 = \frac{v_1}{2} = 0.75A, \quad i_2 = \frac{0 - v_2}{2} = \frac{15.5}{2} = 7.75A.$$

$$\& i_3 = \frac{v_1 - v_2}{4} = \frac{1.5 - (-15.5)}{4} = \frac{17}{4} = 4.25A.$$

② For the below given circuit, find the node voltages, V_1, V_2 & V_3 .



Solⁿ:- At node V_1 , using KCL -

$$6 = \frac{V_1}{5} + \frac{V_1 - V_2}{1} \rightarrow 6 = 1.2V_1 - V_2 \quad \text{--- (1)}$$

at node V_2 - $i_2 = i_3 + i_4$

$$V_1 - V_2 = \frac{V_2}{3} + \frac{V_3}{4}$$

an ideal voltage source, connected betⁿ two nodes forms a super node.

$$\therefore \underline{V_3 - V_2 = 2V} \rightarrow \underline{V_3 = 2V + V_2}$$

$$\therefore V_1 - V_2 = \frac{V_2}{3} + \frac{2 + V_2}{4}$$

$$V_1 = 0.5 + V_2 \left[1 + \frac{1}{3} + \frac{1}{4} \right] \rightarrow V_1 = 0.5 + 1.5833 V_2$$

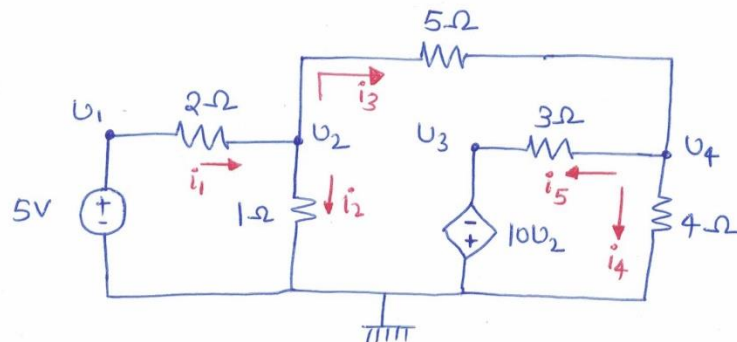
$$V_1 - 1.5833 V_2 = 0.5 \quad \text{--- (2)}$$

on solving ① & ② $V_1 = 10V$, $V_2 = 6V$,

$$\underline{V_3 = 2 + V_2 = 8V.}$$

③

* Find the node voltages v_1, v_2, v_3, v_4 of the circuit shown below.



From fig, $v_1 = 5V$ & $v_3 = -10v_2$

at node v_2 , $i_1 = i_2 + i_3$

$$\frac{5 - v_2}{2} = \frac{v_2}{1} + \frac{v_2 - v_4}{5}$$

$$2.5 = 1.7v_2 - 0.2v_4 \rightarrow 25 = 17v_2 - 2v_4 \quad (1)$$

at node v_4 , $i_3 = i_4 + i_5$

$$\frac{v_2 - v_4}{5} = \frac{v_4}{4} + \frac{v_4 - v_3}{3}$$

$$0 = \frac{v_4}{4} + \frac{v_4}{3} + \frac{10v_2}{3} - \frac{v_2}{5} + \frac{v_4}{5}$$

$$= 15v_4 + 20v_4 + 200v_2 - 12v_2 + 12v_4$$

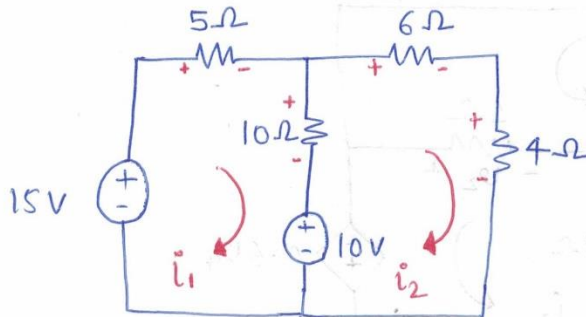
$$= 47v_4 + 188v_2 = 0 \quad (2)$$

Solving (1) & (2), $v_2 = 1V$, $v_4 = -4V$

$$v_3 = -10v_2 = -10V$$

* Mesh Analysis :-

① for the below given circuit, find the loop currents i_1 & i_2 .



using KVL in loop 1.

$$15 - 5i_1 - 10(i_1 - i_2) - 10 = 0$$

$$5 = 15i_1 - 10i_2 \quad \text{--- (1)}$$

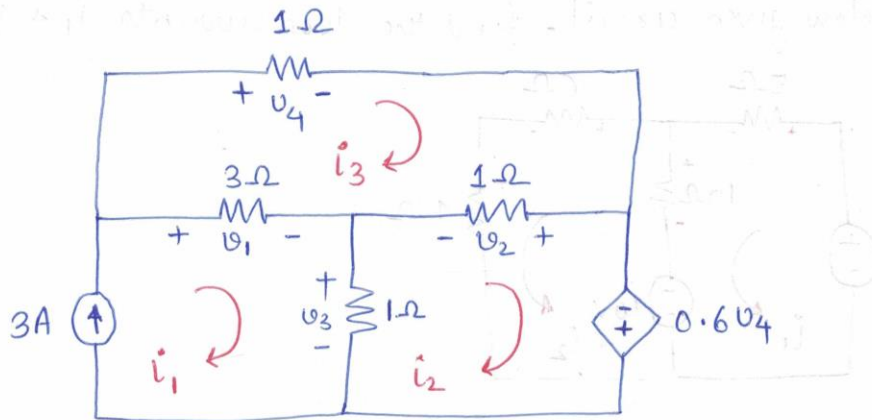
using KVL in loop 2 -

$$10 + 10 \times (i_1 - i_2) - 10i_2 = 0$$

$$10 = -10i_1 + 20i_2 \quad \text{--- (2)}$$

$$\underline{i_1 = 1A} \quad \& \quad \underline{i_2 = 1.0A}$$

② using mesh analysis, determine voltage across all elements.



→ for loop 3: using KVL - $-v_4 - v_2 + v_1 = 0$

using ohm's law - $-1 \times i_3 - 1 \times (i_3 - i_2) + 3(i_1 - i_3) = 0$

$$3i_1 + i_2 - 5i_3 = 0 \quad \text{--- (1)}$$

for loop 2: using KVL - $v_3 + v_2 + 0.6v_4 = 0$

using ohm's law - $1 \times (i_1 - i_2) + 1 \times (i_3 - i_2) + 0.6i_3 = 0$

$$i_1 - 2i_2 + 1.6i_3 = 0 \quad \text{--- (2)}$$

by inspection, $i_1 = 3A$. substituting in (1) & (2)

$$i_2 - 5i_3 = -9 \quad \text{--- (3)}$$

$$-2i_2 + 1.6i_3 = -3 \quad \text{--- (4)}$$

solving (3) & (4) - $i_2 = 3.5A$, $i_3 = 2.5A$.

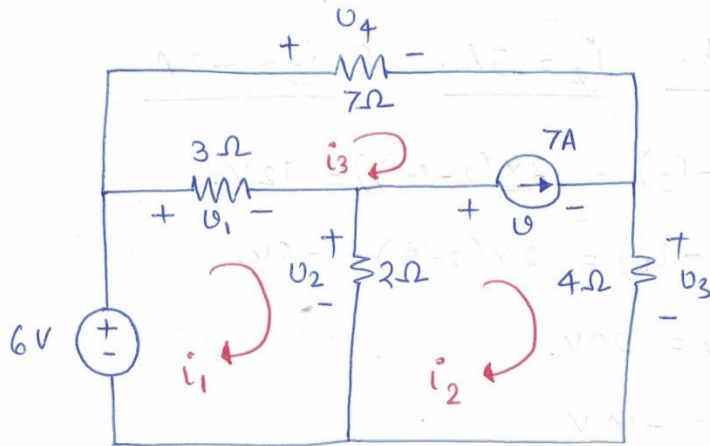
$$v_1 = 3 \times (i_1 - i_3) = 3 \times (3 - 2.5) = 1.5V.$$

$$v_2 = 1 \times (i_3 - i_2) = 1 \times (2.5 - 3.5) = -1V.$$

$$v_3 = 1 \times (i_1 - i_2) = 1 \times (3 - 3.5) = -0.5V$$

$$v_4 = 1 \times i_3 = 2.5V$$

- ③ Determine the voltages across all the elements of the circuit given below, using mesh analysis.



Solution :- using KVL for loop 1 -

$$6 - v_1 - v_2 = 0, \text{ using ohm's law } - 6 = 3(i_1 - i_3) + 2(i_1 - i_2)$$

$$6 = 5i_1 - 2i_2 - 3i_3 \quad \text{--- (1)}$$

for loop 2 using KVL $\rightarrow v_2 - v - v_3 = 0$

$$2(i_1 - i_2) - v - 4i_2 = 0$$

$$v = 2i_1 - 6i_2 \quad \text{--- (2)}$$

for loop 3, using KVL, $-v_4 + v + v_1 = 0$

$$7i_3 + v + 3(i_1 - i_3) = 0$$

$$v = -3i_1 + 10i_3 \quad \text{--- (3)}$$

using (2) & (3), $2i_1 - 6i_2 = -3i_1 + 10i_3$

$$\rightarrow 5i_1 - 6i_2 - 10i_3 = 0 \quad \text{--- (4)}$$

By inspection. $i_1 - i_2 = 7A$

$$5i_1 - 2i_2 - 3i_3 = 6 \quad \text{--- ①}$$

$$5i_1 - 6i_2 - 10i_3 = 0 \quad \text{--- ②}$$

$$\& \quad 0i_1 + i_2 - i_3 = 7 \quad \text{--- ③}$$

Solving ①, ② & ③ -

$$\underline{i_1 = 2A.} \quad \underline{i_2 = 5A.} \quad \& \quad \underline{i_3 = -2A.}$$

$$\therefore v_1 = 3(i_1 - i_3) = 3 \times (2 - (-2)) = 12V.$$

$$v_2 = 2 \times (i_1 - i_2) = 2 \times (2 - 5) = -6V.$$

$$v_3 = 4 \times i_2 = 20V.$$

$$v_4 = 7 \times i_3 = -14V$$

$$v = 2i_1 - 6i_2 = 2 \times 2 - 6 \times 5 = 4 - 30 = \underline{\underline{-26V.}} \quad \therefore \underline{\underline{26V/2.}}$$

$$(i_1 - i_2) \times 3 + (i_2 - i_3) \times 2 = 6 \quad \text{--- nodal eqn at node 1, } 0 = 2.5V - 1V - 1A$$

$$\text{①} - i_1 + i_2 - 2i_3 = 2$$

$$0 = i_1 - i_2 + i_3 \quad \text{--- nodal eqn at node 2, } 5V \text{ node 3}$$

$$3 = i_1 - i_2 - (i_1 - i_2) \times 2$$

$$\text{②} - i_1 + i_2 = 3$$

$$0 = i_1 - i_2 + i_3 \quad \text{--- nodal eqn at node 3, } 5V \text{ node 2}$$

$$0 = (i_1 - i_2) \times 3 + i_3$$

$$\text{③} - 3i_1 + 3i_2 + i_3 = 0$$

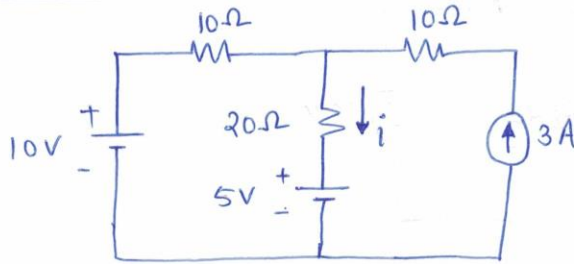
$$3i_1 - 3i_2 - i_3 = 0 \quad \text{--- ④} \quad \text{--- ⑤} \quad \text{--- ⑥}$$

$$\text{②} - 0 = i_1 - i_2 - i_3 \quad \text{--- ⑦}$$

$$\text{--- nodal eqn at node 1, } 5V \text{ node 2}$$

Superposition theorem:-

①



* Find current i using superposition theorem.

Solⁿ:-

using 10V alone, $i_1 = \frac{10}{30} = \frac{1}{3} \text{ A}.$

using 5V alone, $i_2 = -\frac{5}{30} = -\frac{1}{6} \text{ A}.$

using 3A alone, $i_3 = \frac{3 \times 10}{30} = 1 \text{ A}.$

$$\therefore i = i_1 + i_2 + i_3 = \frac{1}{3} - \frac{1}{6} + 1 = \frac{2-1+6}{6} = \frac{7}{6} \text{ A}.$$

→ Prove that superposition theorem cannot be used/applied to compute power dissipated in a resistor due to more than one source.

→ Power dissipated in 20Ω due to 10V alone $= i_1^2 \times 20$
 $= \frac{20}{9} \text{ W}.$

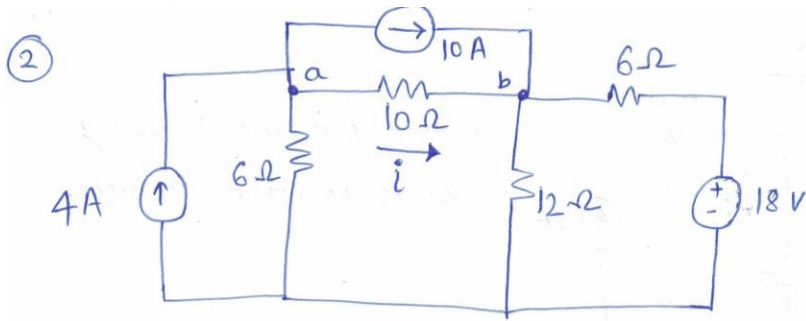
—————||————— 5V alone $= i_2^2 \times 20$
 $= \frac{20}{36} \text{ W}$

—————||————— 3A alone $= 20 \text{ W}.$

Sum of individual power $= \underline{27.77 \text{ W}}.$

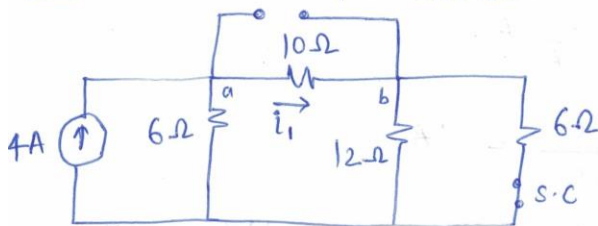
Power dissipated due to current flowing with all 3 sources acting simultaneously $= \left(\frac{7}{6}\right)^2 \times 20 = 27.222 \text{ W}$

Thus, superposition theorem cannot be applied.



Find current i flowing through 10Ω using superposition theorem.

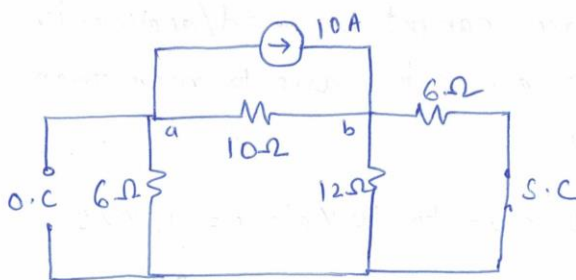
Sol:- i due to 4A alone -



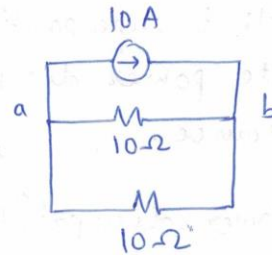
$$R_{eq} = \frac{12 \times 6}{12 + 6} = 4\Omega$$

$$\therefore i_1 = \frac{4 \times 6}{6 + 14} = \frac{6}{5} \text{ A (a to b)}$$

i due to 10A alone -

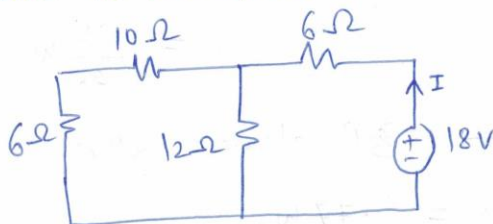


→



$$i_2 = -5 \text{ A (a to b)}$$

i due to 18V alone -



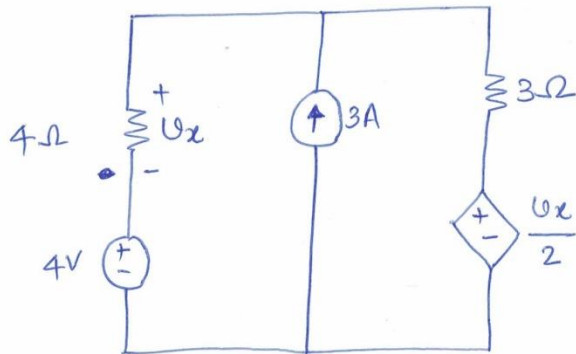
$$I = \frac{18}{6 + 12 \parallel 12} = \frac{18}{6 + 6.857} = 1.4 \text{ A}$$

$$I_3 = \frac{1.4 \times 12}{28} = \frac{12}{20} = 0.6 \text{ A (b to a)}$$

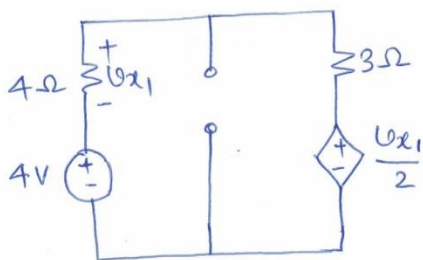
$$\text{Net current } i = i_1 + i_2 + i_3$$

$$= \frac{6}{5} - 5 - 0.6 = -4.4 \text{ A (from a to b)}$$

- ③ Find the value of voltage ' V_x ' in the below given circuit using superposition theorem.



Sol?- Considering 4V source alone -



using KVL -

$$4 - 4 \times I - 3I - \frac{V_{x1}}{2} = 0$$

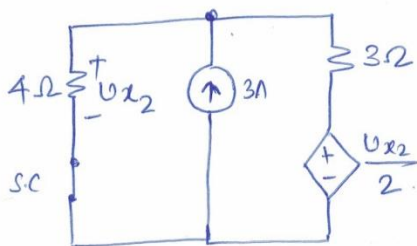
$$V_{x1} = -4 \times I$$

$$\therefore 4 - 4I - 3I - \frac{(-4I)}{2} = 0$$

$$4 = 7I - 2I = 5I \quad I = \frac{4}{5} = 0.8A$$

$$\therefore V_{x1} = -4 \times I = -3.2V$$

Considering 3A source alone -



using KCL -

$$3 = \frac{V_{x2}}{4} + \frac{V_{x2} - \frac{V_{x2}}{2}}{3}$$

$$3 = \frac{5V_{x2}}{12}$$

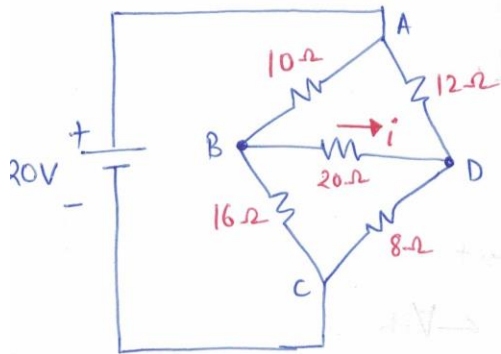
$$V_{x2} = \frac{36}{5} V$$

$$\therefore V_x = V_{x1} + V_{x2}$$

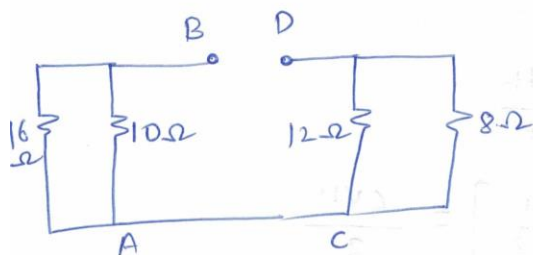
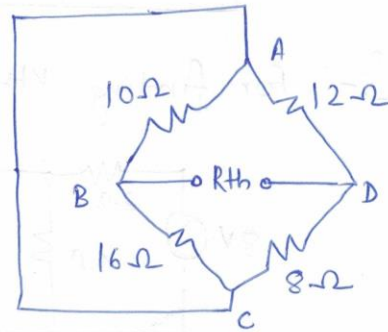
$$= -3.2 + \frac{36}{5} = \underline{\underline{4V}}$$

Thevenin's theorem:-

- ① Find current i using Thevenin's theorem for the following circuit.



Solⁿ:- For finding R_{th} , voltage source is ^{made} short-circuit.

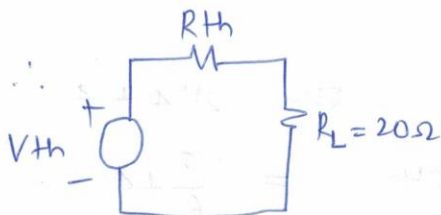


$$R_{th} = (16 \parallel 10) + (12 \parallel 8)$$

$$R_{th} = 10.954 \Omega$$

for V_{th} - $V_B - V_D = V_{th}$.

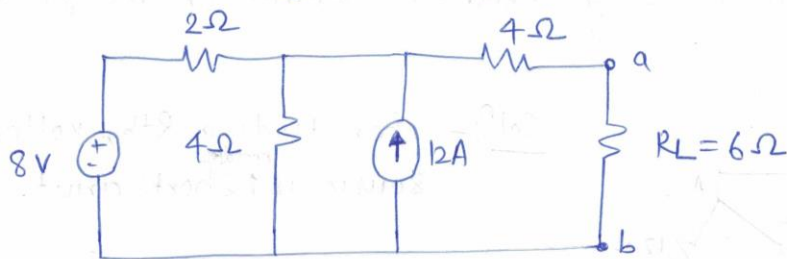
$$V_{th} = \frac{16 \times 20}{26} - \frac{8 \times 20}{20} = \underline{\underline{4.308 \text{ V}}}$$



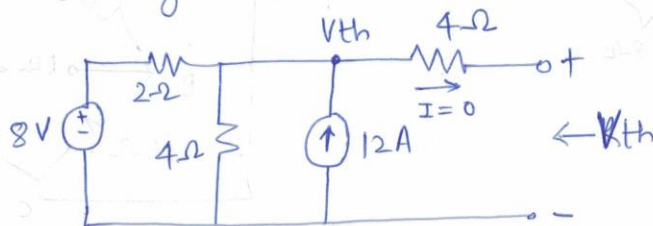
$$I = \frac{4.308}{10.954 + 20}$$

$$I = 0.139 \text{ A}$$

② Find current through $R_L = 6\Omega$ for the circuit shown below -



Sol:- For finding V_{th} , Remove R_L



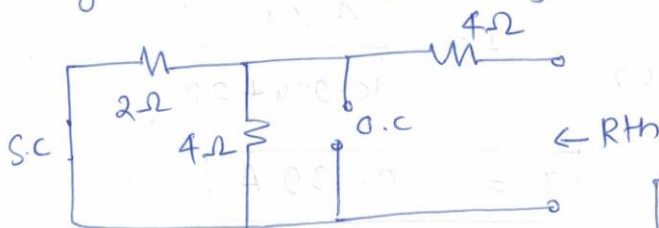
Using nodal analysis -

$$\frac{0 - V_{th}}{2} + 12 = \frac{V_{th}}{4}$$

$$16 = V_{th} \left[\frac{1}{4} + \frac{1}{2} \right] = \frac{6V_{th}}{8}$$

$$\therefore V_{th} = \frac{16 \times 8}{6} = \frac{128}{3} = 21.333 \text{ V.}$$

For finding R_{th} , Replace voltage source & current source by s.c & o.c respectively.



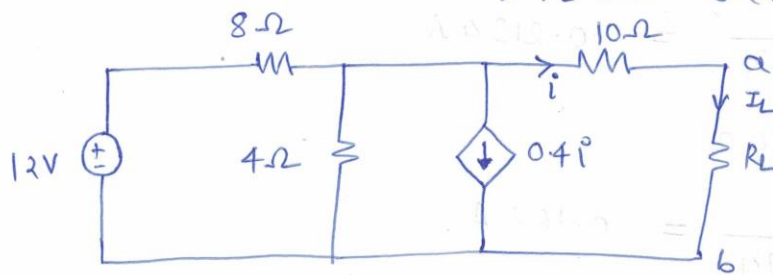
$$R_{th} = 2 \parallel 4 + 4$$

$$= \frac{8}{6} + 4$$

$$R_{th} = \frac{32}{6} \Omega$$

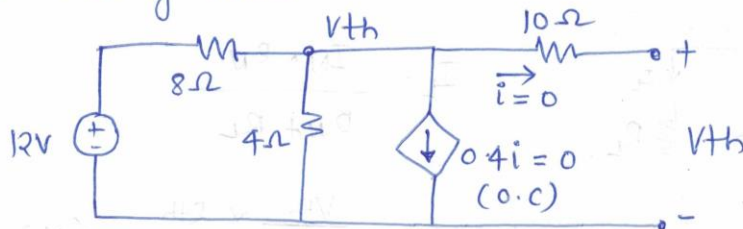
$$I = \frac{V_{th}}{R_{th} + R_L} = \frac{21.333}{\frac{32}{6} + 6} = \underline{\underline{1.882 \text{ A.}}}$$

- ③ For the circuit shown below, find load current I_L when
 i) $R_L = 5\Omega$ & (ii) $R_L = 10\Omega$.



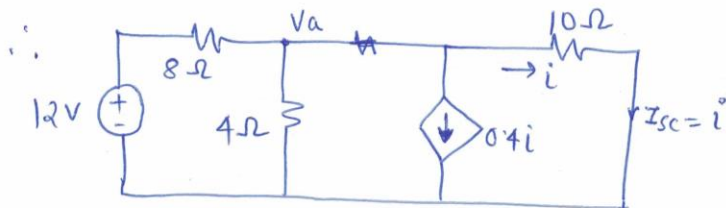
~~For finding R_{th} , replace ideal voltage source by S.C. Do not alter dependent sources.~~

For finding V_{th} -



$$\therefore V_{th} = \frac{4 \times 12}{12} = 4V$$

If network contains any dependent source, then R_{th} can be computed using $R_{th} = \frac{V_{th}}{I_{sc}}$ where I_{sc} is the short circuit current.



using Nodal analysis,

$$\frac{12 - V_a}{8} = \frac{V_a}{4} + 0.4i + \frac{V_a}{10}$$

$$\text{here, } i = \frac{V_a}{10}$$

$$\therefore \frac{12 - 10i}{8} = \frac{10i}{4} + 0.4i + i$$

$$\rightarrow 1.5 = 3.9i + 1.25i = 5.15i \rightarrow \underline{\underline{i = I_{sc} = 0.2912A}}$$

$$\therefore R_{th} = \frac{V_{th}}{I_{sc}} = \frac{4}{0.2912} = \underline{\underline{13.74\Omega}}$$

I_L when $R_L = 5\Omega$ -

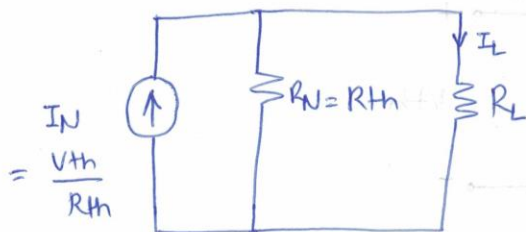
$$I_L = \frac{4}{13.74 + 5} = 0.2134 \text{ A.}$$

I_L when $R_L = 10\Omega$

$$I_L = \frac{4}{23.74} = 0.168 \text{ A.}$$

* For Norton's Theorem -

Find V_{th} & R_{th} . $I_N = \frac{V_{th}}{R_{th}}$, $R_N = R_{th}$.



$$I_L = \frac{I_N \cdot R_N}{R_N + R_L}$$

$$= \frac{\frac{V_{th}}{R_{th}} \times R_{th}}{R_{th} + R_L} \quad (\because R_N = R_{th})$$

$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

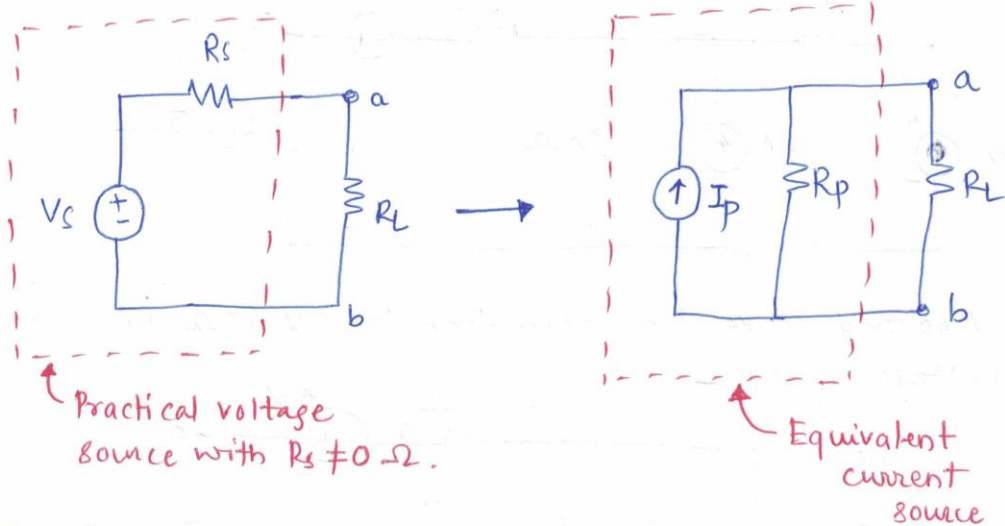


Assuming 20Ω is

$$20\Omega \parallel 10\Omega = \frac{20 \times 10}{20 + 10} = \frac{200}{30} = 6.67\Omega$$

* Source Transformation:-

* Applicable only to practical voltage & current source.



Cond:- The open circuit voltage across a-b terminals must be same for both cases

i.e. $I_p \times R_p = V_s$ (as current through $R_s = 0$).

$$\therefore I_p = \frac{V_s}{R_p} \quad - (1)$$

→ The short circuit current for both cases must be same.

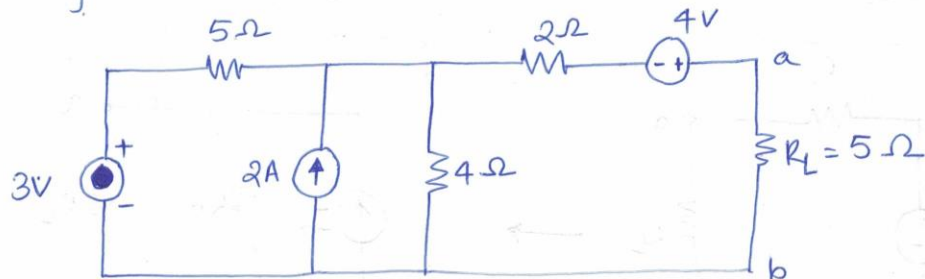
$$\text{i.e. } \frac{V_s}{R_s} = I_p \quad - (2)$$

Equating ① & ② gives $R_p = R_s$

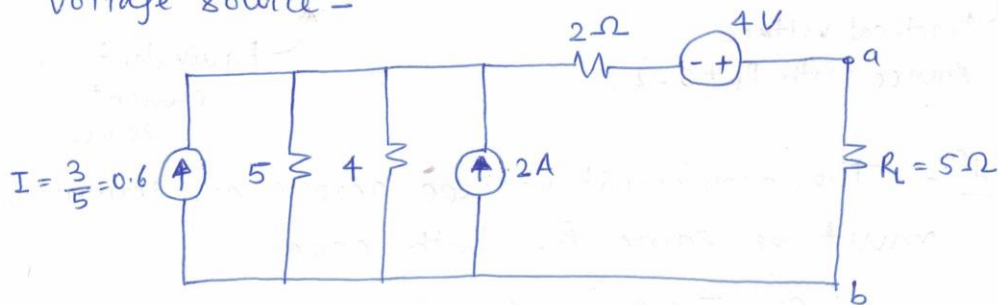
$$\& \quad \underline{\underline{I_p = \frac{V_s}{R_s}}}$$

Norton's Theorem:-

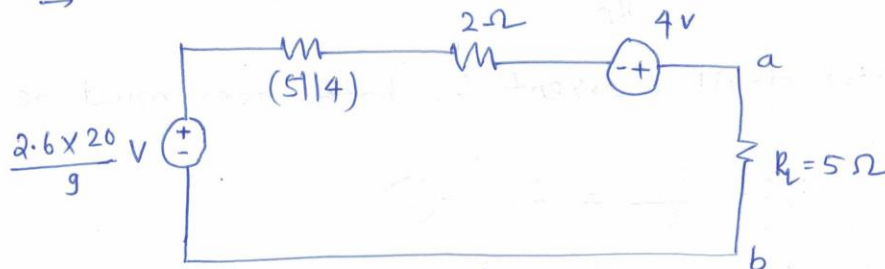
- ① Using source transformation & Norton's theorem, find voltage across $R_L = 5\Omega$.



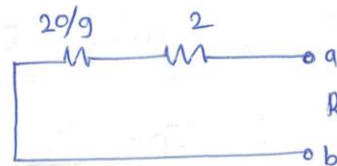
Solⁿ:- Applying source transformation to 3V & 5Ω practical voltage source -



parallel current sources can be added. Modified circuit after source transformation will be -

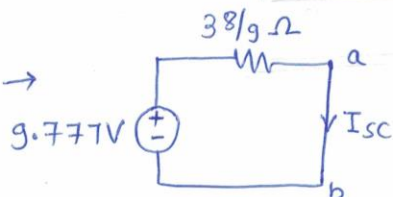


for $R_N = R_{Th}$ -

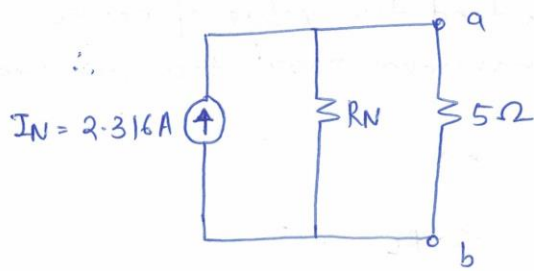


$$R_N = R_{Th} = \frac{20}{9} + 2 = \frac{38}{9} \Omega$$

for I_{sc} →



$$I_{sc} = \frac{9.777}{38/9} = \underline{\underline{2.316 A}} = I_N$$

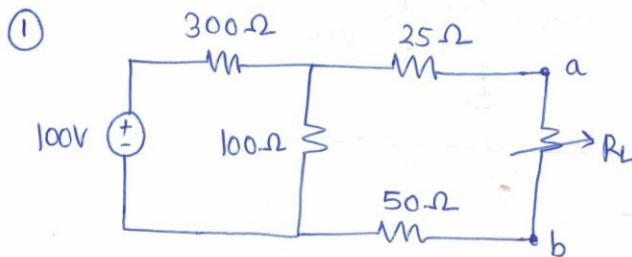


$$I_{5\Omega} = \frac{2.316 \times \frac{38}{9}}{\frac{38}{9} + 5}$$

$$I_{5\Omega} = 1.06 \text{ A.}$$

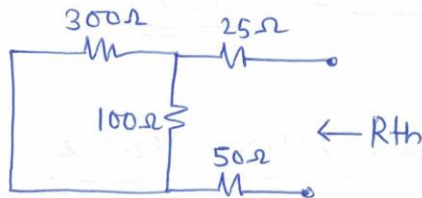
$$\begin{aligned} \text{Voltage across } 5\Omega &= 1.06 \times 5 \\ &= \underline{\underline{5.30 \text{ V}}} \end{aligned}$$

* Maximum Power Transfer Theorem:-



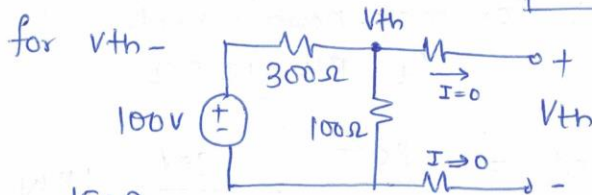
- * Find the value of R_L for maximum power transfer.
- * Calculate power transferred under max. power transfer condition.

Solⁿ:- For maximum power transfer, $R_L = R_{th}$ seen from load terminals a-b.



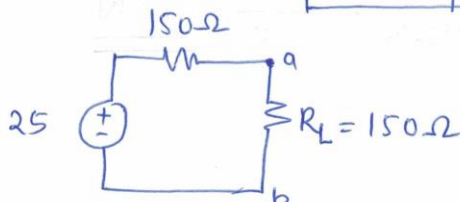
$$\begin{aligned} R_{th} &= (300 \parallel 100) + 25 + 50 \\ &= \frac{300 \times 100}{400} + 75 \end{aligned}$$

$$\boxed{R_{th} = 150 \Omega.}$$



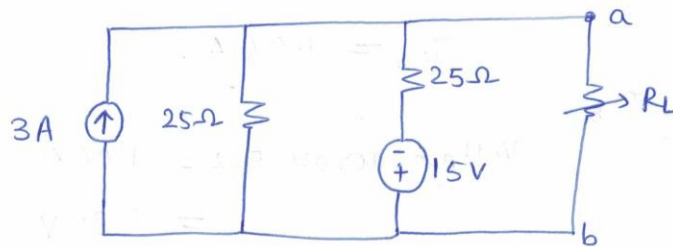
$$\therefore V_{th} = \frac{100 \times 100}{400}$$

$$\boxed{V_{th} = 25 \text{ V}}$$

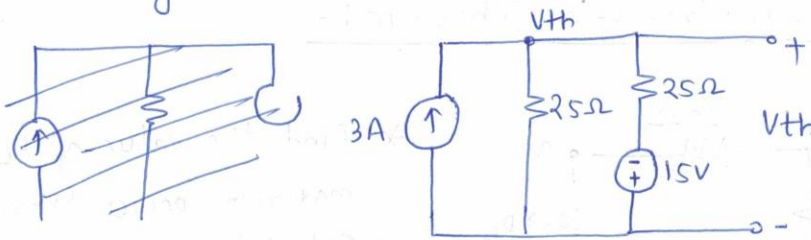


$$\begin{aligned} \text{Power dissipated in } R_L &= 150\Omega \\ &= \frac{V_{th}^2}{4R_L} = \frac{25^2}{4 \times 150} \end{aligned}$$

- ② For the network shown below, find the value of power delivered to the load under maximum power transfer cond.



Solⁿ:- Applying Thevenin's theorem to find V_{th} -

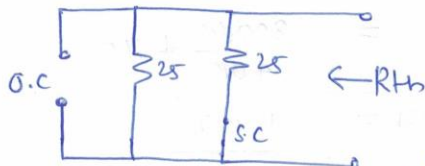


using nodal analysis -

$$3 = \frac{V_{th}}{25} + \frac{V_{th} - (-15)}{25} \Rightarrow 3 = \frac{2V_{th}}{25} + \frac{15}{25}$$

$$\therefore V_{th} = \frac{2.4 \times 25}{2} = \underline{\underline{30.0V}}$$

for R_{th} -



$$R_{th} = 25 \parallel 25 = 12.5\Omega$$

$\therefore$ For max. power transfer

$$R_L = R_{th} = 12.5\Omega$$

$$\therefore \text{Power transferred} = \frac{30^2}{4 \times 12.5} = \frac{900}{50} = \underline{\underline{18W}}$$